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A Models and Implementation

A.1 Entropy

The entropy of the output distribution is implemented as follows,

$$\mathcal{H}_{ij} = 2^{-\sum_{\tilde{w}\in\mathcal{W}} p_{ij}(\tilde{w}) \log_2 p_{ij}(\tilde{w})} \quad (13)$$

where $\mathcal{W}$ is the set of all possible words in the vocabulary.

A.2 Proxy LLMs

The proxy LLMs considered are LLaMA-{7B, 13B, 30B} (Touvron et al., 2023), OPT-{125m, 1.3B, 13B, 30B} (Zhang et al., 2022), GPT-J-6B (Wang and Komatsuzaki, 2021) and GPT-NeoX-20B (Black et al., 2022).

A.3 SelfCheckGPT's Systems

Question Answering: The generation systems G1 and G2 are T5-Large fine-tuned to SQuAD (Rajpurkar et al., 2016) and RACE (Lai et al., 2017), respectively. The answering system A is Longformer (Beltagy et al., 2020) fine-tuned to the RACE dataset. The answerability system U is also Longformer, but fine-tuned to SQuAD2.0.

LLM for Prompting: We consider two LLMs, GPT-3 (text-davinci-003) and ChatGPT (gpt-3.5-turbo) We note that during the data creation and annotation, GPT-3 (text-davinci-003) was the state-of-the-art LLM available; hence, GPT-3 was used as the main LLM generating WikiBio passages.

B SelfCheckGPT with QA

Previous work showed that implementing question generation (in Equation 2) with two generators (G1 generates the question and associated answer, and G2 generates distractors) yields higher-quality distractors (Manakul et al., 2023). Thus, a two-stage generation is adopted in this work as follows:

$$q, a \sim P_{\texttt{G1}}(q, a|r_i); \;\; \mathbf{o}_{\setminus a} \sim P_{\texttt{G2}}(\mathbf{o}_{\setminus a}|q, a, R) \quad (14)$$

where $\mathbf{o} = \{a, \mathbf{o}_{\setminus a}\} = \{o_1, ..., o_4\}$. In addition, to filter out bad (unanswerable) questions, we define an answerability score (Raina and Gales, 2022):

$$\alpha = P_{\texttt{U}}(\text{answerable}|q, \text{context}) \quad (15)$$

where the context is either the response R or sampled passages S^n, and $\alpha \to 0.0$ for unanswerable and $\alpha \to 1.0$ for answerable. We use α to filter out unanswerable questions which have α lower than a threshold. Next, we derive how Bayes' theorem can be applied to take into account the number of answerable/unanswerable questions.

B.1 SelfCheckGPT-QA with Bayes

Let $P(\text{F})$ denote the probability of the i-th sentence being non-factual, and $P(\text{T})$ denote the probability of the i-th sentence being factual. For a question q, the probability of i-th sentence being non-factual given a set of matched answers $L_{\texttt{m}}$ and a set of not-matched answers $L_{\texttt{n}}$ is:

$$\begin{aligned} &P(\text{F}|L_{\texttt{m}}, L_{\texttt{n}}) \\ &= \frac{P(L_{\texttt{m}}, L_{\texttt{n}}|\text{F})P(\text{F})}{P(L_{\texttt{m}}, L_{\texttt{n}}|\text{F})P(\text{F}) + P(L_{\texttt{m}}, L_{\texttt{n}}|\text{T})P(\text{T})} \\ &= \frac{P(L_{\texttt{m}}, L_{\texttt{n}}|\text{F})}{P(L_{\texttt{m}}, L_{\texttt{n}}|\text{F}) + P(L_{\texttt{m}}, L_{\texttt{n}}|\text{T})} \end{aligned} \quad (16)$$

where we assume the sentence is equally likely to be False or True, i.e. $P(\text{F}) = P(\text{T})$. The probability of observing $L_{\texttt{m}}, L_{\texttt{n}}$ when the sentence is False (non-factual):

$$\begin{aligned} &P(L_{\texttt{m}}, L_{\texttt{n}}|\text{F}) \\ &= \prod_{a\in L_{\texttt{m}}} P(a = a_R|F) \prod_{a'\in L_{\texttt{n}}} P(a' \neq a_R|F) \\ &= (1-\beta_1)^{N_{\texttt{m}}}(\beta_1)^{N_{\texttt{n}}} \end{aligned} \quad (17)$$

and probability of observing $L_{\texttt{m}}, L_{\texttt{n}}$ when the sentence is True (factual):

$$\begin{aligned} &P(L_{\texttt{m}}, L_{\texttt{n}}|\text{T}) \\ &= \prod_{a\in L_{\texttt{m}}} P(a = a_r|T) \prod_{a'\in L_{\texttt{n}}} P(a' \neq a_r|T) \\ &= (\beta_2)^{N_{\texttt{m}}}(1-\beta_2)^{N_{\texttt{n}}} \end{aligned} \quad (18)$$

where $N_{\texttt{m}}$ and $N_{\texttt{n}}$ are the number of matched answers and the number of not-matched answers, respectively. Hence, we can simplify Equation 16:

$$P(\text{F}|L_{\texttt{m}}, L_{\texttt{n}}) = \frac{\gamma_2^{N_{\texttt{n}}}}{\gamma_1^{N_{\texttt{m}}} + \gamma_2^{N_{\texttt{n}}}} \quad (19)$$

where $\gamma_1 = \frac{\beta_2}{1-\beta_1}$ and $\gamma_2 = \frac{\beta_1}{1-\beta_2}$. Lastly, instead of rejecting samples having an answerability score below a threshold,[7] we find empirically that soft-counting (defined below) improves the detection performance. We set both β_1 and β_2 to 0.8.

[7] α is between 0.0 (unanswerable) and 1.0 (answerable). Standard-counting $N_{\texttt{m}}$ and $N_{\texttt{n}}$ can be considered as a special case of soft-counting where α is set to 1.0 if α is greater than the answerability threshold and otherwise α is 0.0.